

v^0 $t=0$
 v^1 $t=T$

$$w(t) = at^3 + bt^2 + ct + d$$



$$v(t) = 3at^2 + 2bt + c$$

1. $t = 0, w(0) = w^0$

2. $t = T, w(T) = w^1$

3. $t = 0, v(0) = v^0$

4. $t = T, v(T) = v^1$

$$w(0) = d = w^0$$

$$w(T) = aT^3 + bT^2 + cT + d = w^1$$

$$v(0) = c = v^0$$

$$v(T) = 3aT^2 + 2bT + c = v^1$$

Solving the equations